MsImpute: Estimation of Missing Peptide Intensity Values in Label-Free Mass Spectrometry

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In Brief

In shotgun proteomics, peptide precursors are quantified by MS1 and identified by sequencing (fragmentation) in MS2 or PASEF. The stochastic selection of peptides at MS1 scanning step, or poor MS2 signals, prevent consistent quantification of peptides and proteins across samples, and result in missing values. We are interested in estmation of missing peptide intensity values by imputation or Peptide Identity Propagation (PIP) between runs based on similarity of MS1 measurements in label-free mass spectrometry data (both LC-MS and TIMS-LC-MS). We developed *msImpute*, which offers a number of workflows for *imputation* of intensity data under MAR and MNAR assumptions, as well as estimation of missing values by *feature-based PIP*. The package also offers diagnostic tools for evaluation of imputation.

A primer on missing values in mass spectrometry

Liquid-chromatography-coupled tandem mass spectrometry (LC-MS/MS) is the leading technology for quantitative analysis of proteins expressed in complex biological samples. Proteins in cell or tissue lysates are quantified by first digesting them into peptides. Peptides are separated using liquid chromatography column and then interfaced the source of the mass spectrometer where they are ionized and converted to gas phase. The separated and ionized peptide precursors are subjected to mass analysis in mass spectrometer. More recently, high resolution Ion Mobility (IM) is used as another dimension for peptide separation. The mass, charge, retention time and IM collectively can define the identity of peptides existing in a sample. It is known that more than 100,000 detectable peptide species elute in single shotgun proteomics runs (Michalski et al., 2011). The mass spectrometer, however, only selects a small subset (usually 10) most abundant peptides for sequencing at each MS1 survey scan in a run (Run is a technical term for sample in mass spectrometry. A peptide is sequenced by fragmentation. For simplicity we use the term sequencing in this work.). This compromises consistent quantification of peptides across runs, as the sets of peptide precursor ions selected for sequencing could be different between runs.

The low sampling efficiency and stochastic nature of Top N precursor selection in MS1 limits depth of protein coverage in large clinical cohorts and hinders quantification of low abundance ions in complex samples, leading to the prevalent problem of missing values. When a peptide is sequenced in a run, its MS1 attributes such as m/z, retention time and IM are available to the experimenter. Data completeness can, hence, be improved by transferring peptide identities from runs where the peptide is identified by sequencing to runs where it is not sequenced, but is detected at the MS1 level based on the similarity of the MS1 attributes.

To alleviate the missing values problem in mass spectrometry data, peptide identifications are matched between runs to increase the number of peptides and proteins quantified across runs. The existing PIP approaches transfer sequence information to a run if a peptide-like signal is present for which the MS1 measurements are within a pre-defined tolerance of MS1 attributes of a peptide identified by sequencing. Some approaches transfer identifications from the run with largest number of identified peptide sequences, or from top k correlated runs. Therefore, the accuracy of the existing approaches is limited by the selection of runs used as reference for PIP and the specified tolerance. These approaches are also inherently limited by the lack of probability measure to assign confidence and filter likely false positive results. We propose to a new PIP approach that leverages output tables from MaxQuant, and obviates the reliance of PIP on arbitrary tolerance thresholds and choices of reference runs.

An alternative solution to missing values is imputation. Missing values can occure at random (MAR) during data processing by incorrect peak matching, as an example. In complex samples, the low abundance of a peptide can be a limiting factor for quantification of the peptide, resulting in Missing Not at Random (MNAR) missingness. We demonstrate that msImpute has the capability to correctly estimate the missing intensity values, imporve coverage and data completeness, and maintain low false discoveries in differential abundance studies.

Find ***msImpute*** on Bioconductor, or see

https://github.com/DavisLaboratory/msImpute

for latest developments.

What does msImpute do?

MsImpute is a R package for estimation of missing peptide intensity in label-free proteomics experiments. The missing values are imputed by low-rank approximation of the underlying data matrix if they are MAR (method = "v2"), by Barycenter approach if missingness is MNAR ("v2-mnar"), or by Peptide Identity Propagation (PIP). While "v2" approach is more appropriate for imputation of data acquired by DIA, "v2-mnar" is designed for imputation of DDA, TMT and time-series datasets. However, the true dynamic range can not be reliably recovered by imputation, particularly in datasets with small sample sizes (for example, 3-5 replicates per experimental condition).

Our PIP approach infers the missing intensity values for an identification based on similarity of LC-MS features of peptide-like signals detected in MS1 (e.g. by a feature detector) and the identified peptides. We currently support MaxQuant outputs, including DDA-PASEF datasets. Our PIP enhances data completeness, while reporting weights that measure the confidence in propagation.

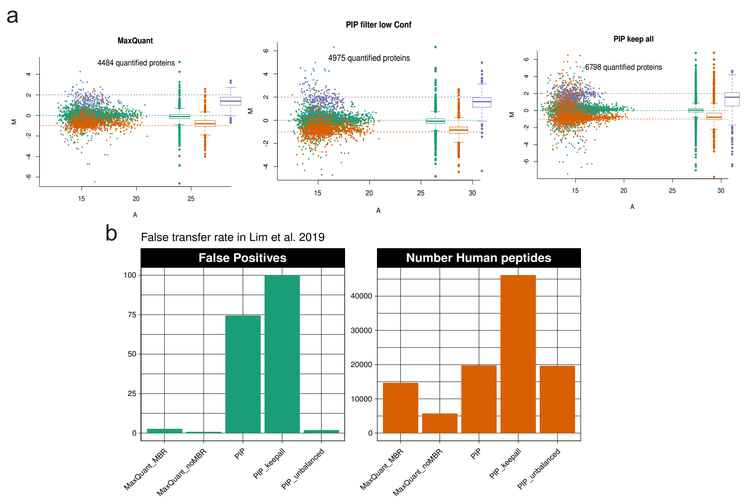

Figure 1: The PIP framework in msImpute can increase depth of coverage, while maintaining reasonable False Transfer Rate. *The PIP framework returns a confidence score for every intensity value estimated by propagation of peptide identifications. The user can select to only retain PIP with confidence score above a threshold, or retain all propagations. a) is protein coverage before and after PIP in a DDA-PASEF dataset containing a hybrid Yeast (Green), HeLa (Orange), Ecoli (Purple) proteome. b) False transfer rate in Lim et al. 2019, which consists of 20 HeLa and Yeast runs and 20 HeLa-only runs. If PIP is carried out so that identifications are propagated from all runs in the dataset, msimpute PIP workflow does detect ~5000 more peptides than MaxQuant MBR, but this comes at the cost of detecting 75% of the Yeast identifications in HeLa runs (hence False Positives). If we only restrict identifications from HeLa runs to be transferred (i.e. unbalanced PIP), the False Positive rate will be lower than MaxQuant MBR. In real scenarios, the performance of the PIP workflow will be close to PIP unbalanced results presented here.*

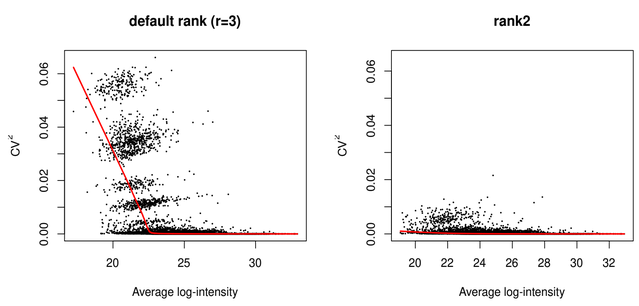

Figure 2: Imputation by low-rank approximation and distribution averaging. *MsImpute v2 uses weighted average of two distributions to impute missing peptide intensity: distribution of the data under the low-rank model and data distribution imputed by the Down-Shift approach (for not at random missingness). The rank is estimated automatically by the method from the data. However, for datasets with a total of 6-8 runs, we recommend rank-2 models. The users can use the mean - squared coefficient of variation plots (CV) as a diagnostic to compare imputed data under different ranks and even different imputation algorithms.*

How does msImpute perform?

We assessed the performance of the barycenter imputation algorithm and the PIP framework introduced in v2 in the context of Differential Abundance analysis. We selected three published DDA datasets, which contain UPS1 (PXD002370) or other proteins (PAS00589) spiked into a constant background. The spiked proteins are the true positives, and any peptides from background proteins detected as differentially abundant is considered a false discovery, or false positive. The dataset in figure 3a consists of 8 experimental group and 24 runs (3 replicates per group). Datasets in 3b and 3c consist of two experimental group and 6 runs (also 3 replicates per group). Figure 3 demonstrates that the barycenter approach outperforms State-of-the-art (SOTA) imputation algorithms in both small (n=6 runs) and large (n=24 runs) datasets. The figures also demonstrates that if we retain all peptide intensity values estimated by PIP, but down-weight them according to confidence scores (hence PIP_weight), we expect results that are either as good, or better than imputation by SOTA algorithms.

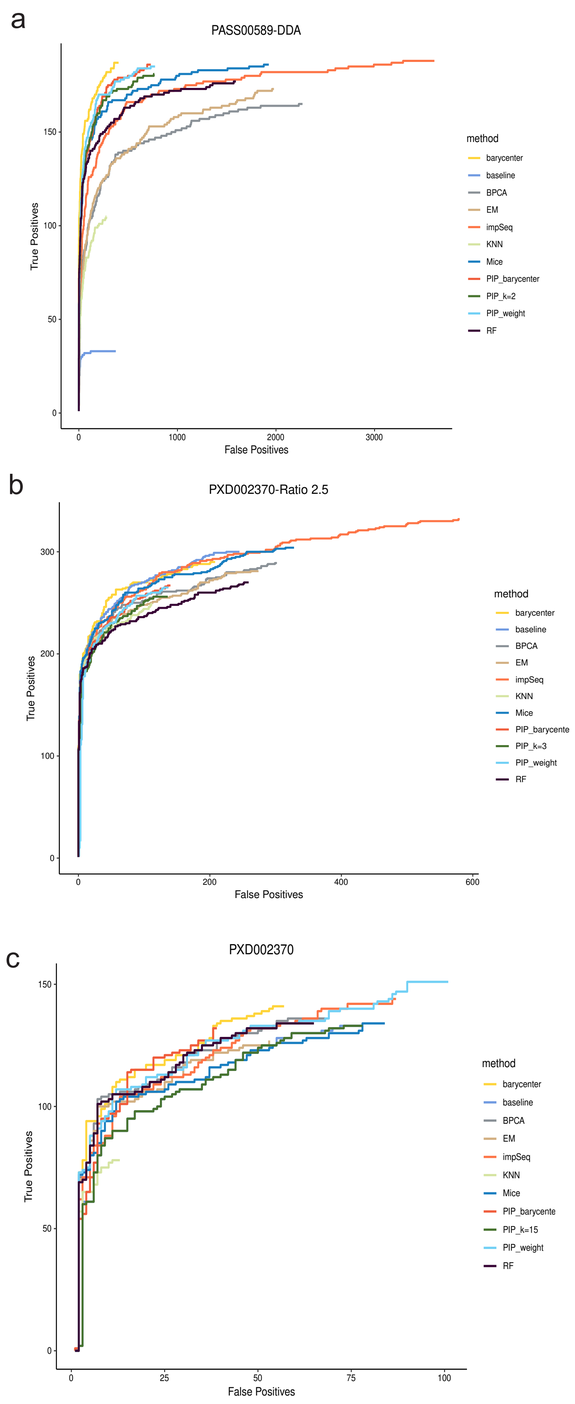

Figure 3: ROC curve for Differential Abundance in spiked-in DDA datasets. *The larger the area under the curve, the better. The length of the curves is proportional to the number of differentially abundant peptides detected at FDR 0.05. Differential abundance tests are done in limma at the peptide level. Barycenter is implemented in* "v2-mnar" method of msImpute imputation function. PIP refers to intensity estimation by PIP, where only estimates with confidence score larger than 0.5 are retained. In PIP-weight, all inferred intensity values are retained, but weighted according to their confidence score in limma linear models.

Reference

Hediyeh-zadeh, S., Webb, A. I., & Davis, M. J. (2020). MSImpute: Imputation of label-free mass spectrometry peptides by low-rank approximation. bioRxiv.